

ranges::to: A function to convert any range to a container

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Abstract

We propose a function to copy or materialize any range (containers and views alike) to a container.

Revisions

Revision 3

- Add support for `from_range_t`
- Add support for nested containers
- Remove syntax without parenthesis

Revision 2

- Remove the implicit const removal when converting an associative container to a container of pairs
- Use CTAD to determine the value type of the returned container
- Attempt at wording

Revision 1

- Split out the proposed constructors for string view and span into separate papers ([[P1391](#)] and [[P1394](#)] respectively)
- Use a function based approach rather than adding a constructor to standard containers, as it proved unworkable.

Quick Overview

We propose all the following syntaxes to be valid constructs

```
std::list<int> l;  
std::map<int, int> m;  
  
// copy a list to a vector of the same type  
same_as<std::vector<int>> auto a = ranges::to<std::vector<int>>(l);  
//Specify an allocator  
same_as<std::vector<int, Alloc>> auto b = ranges::to<std::vector<int, Alloc>(l, alloc);  
// copy a list to a vector of the same type, deducing value_type  
same_as<std::vector<int>> auto c = ranges::to<std::vector>(l);  
// copy to a container of types ConvertibleTo  
same_as<std::vector<long>> auto d = ranges::to<std::vector<long>>(l);  
  
//Supports converting associative container to sequence containers  
same_as<std::vector<std::pair<const int, int>>>  
    auto f = ranges::to<vector<std::pair<const int, int>>>(m);  
  
//Supports converting sequence containers to associative ones  
same_as<std::map<int, int>> auto g = f | ranges::to<map>();  
  
//Pipe syntaxe  
same_as<std::vector<int>> auto g = l | ranges::view::take(42) | ranges::to<std::vector>();  
  
//Pipe syntax with allocator  
auto h = l | ranges::view::take(42) | ranges::to<std::vector>(alloc);  
  
//The pipe syntax also support specifying the type and conversions  
auto i = l | ranges::view::take(42) | ranges::to<std::vector<long>>();  
  
// Nested ranges  
std::list<std::forward_list<int>> lst = {{0, 1, 2, 3}, {4, 5, 6, 7}};  
auto vec1 = rangesnext::to<std::vector<std::vector<int>>>(lst);  
auto vec2 = rangesnext::to<std::vector<std::deque<double>>>(lst);
```

Tony tables

Before	After
<pre>std::list<int> lst = /*...*/; std::vector<int> vec {std::begin(lst), std::end(lst)};</pre>	<pre>std::vector<int> vec = lst ranges::to<std::vector>();</pre>
<pre>auto view = ranges::iota(42); vector < iter_value_t< iterator_t<decltype(view)> > > vec; if constexpr(SizedRanged<decltype(view)>) { vec.reserve(ranges::size(view)); } ranges::copy(view, std::back_inserter(vec));</pre>	<pre>auto vec = ranges::iota(42) ranges::to<std::vector>();</pre>
<pre>std::map<int, widget> map = get_widgets_map(); std::vector< typename decltype(map)::value_type > vec; vec.reserve(map.size()); ranges::move(map, std::back_inserter(vec));</pre>	<pre>auto vec = get_widgets_map() ranges::to<vector>();</pre>

Motivation

Most containers of the standard library provide a constructors taking a pair of iterators.

```
std::list<int> lst;
std::vector<int> vec{std::begin(lst), std::end(lst)};
//equivalent too
std::vector<int> vec;
std::copy(it, end, std::back_inserter(vec));
```

While, this feature is very useful, as converting from one container type to another is a frequent use-case, it can be greatly improved by taking full advantage of the notions and tools offered by ranges.

Indeed, given all containers are ranges (ie: an iterator-sentinel pair) the above example can be rewritten, without semantic as:

```
std::list<int> lst;
```

```
std::vector<int> vec = lst | ranges::to<std::vector>();
```

The above example is a common pattern as it is frequently preferable to copy the content of a `std::list` to a `std::vector` before feeding it an algorithm and then copying it back to a `std::vector`.

As all containers and views are ranges, it is logical they can themselves be easily built out of ranges.

View Materialization

The main motivation for this proposal is what is colloquially called *view materialization*. A view can generate its elements lazily (upon increment or decrement), such as the value at a given position of the sequence iterated over only exist transiently in memory if an iterator is pointing to that position. (Note: while all lazy ranges are views, not all views are lazy).

View materialization consists in committing all the elements of such view in memory by putting them into a container.

The following code iterates over the numbers 0 to 1023 but only one number actually exists in memory at any given time.

```
std::iota_view v{0, 1024};
for (auto i : v) {
    std::cout << i << ' ';
}
```

While this offers great performance and reduced memory footprint, it is often necessary to put the result of the transformation operated by the view into memory. The facilities provided by [\[P0896R3\]](#) allow to do that in the following way:

```
std::iota_view v{0, 1024};
std::vector<int> materialized;
std::copy(v, std::back_inserter(materialized));
```

This proposal allows rewriting the above snippet as:

```
auto materialized = std::iota_view{0, 1024} | std::ranges::to<std::vector>();
```

Perhaps the most important aspect of view materialization is that it allows simple code such as:

```
namespace std {
    split_view<std::string_view> split(std::string_view);
}
auto res = std::split("Splitting strings made easy")
    | std::ranges::to<std::vector>();
```

Indeed, a function such as `split` is notoriously hard to standardize ([P0540], [N3593]), because without lazy views and `std::string_view`, it has to allocate or expose an expert-friendly interface. The view materialization pattern further let the *caller* choose the best container and allocation strategy for their use case (or to never materialize the view should it not be necessary). And while it would not make sense for a standard-library function to split a string into a vector it would allocate, it's totally reasonable for most applications to do so.

This paper does not propose to standardize such `split` function - a `split_view` exist in [P0896R3], however, view materialization is something the SG-16 working group is interested in. Indeed, they have considered APIs that could rely heavily on this idiom, as it has proven a natural way to handle the numerous ways to iterate over Unicode text. Similar ideas have been presented in [P1004].

```
auto sentences =
    text(blob)
    normalize<text::nfc> |
    graphemes_view |
    split<sentences> | ranges::to<std::vector<std::u8string>>();
```

Design

Conceptually, `to` is a function template with multiple overloads:

- One that is templated on a container type and convert a range to that type using the most efficient method depending on the type of that container.
- One that accepts a container as template parameter and deduced the value type of that container using CTAD
- One that offers a pipe adaptor object over both these overloads

Care was taken to support move only iterators, ranges of ranges, non const views, associative containers (in either direction)

Should `ranges::to` be able to call `reserve/assigned` ?

For performance reasons, `ranges::to` should be able to reserve memory in the container before assigning the range whenever possible. This cannot be done in the constructor of individual ranges because a range meeting the requirements of `SizedRange` doesn't imply that the distance between each of the iterator forming the range can be computed efficiently, if at all.

Both `cmcstl2` and `ranges-v3` determine whether or not a container can be reserved and assigned by the mean of a `ReserveAndAssignable` concept that is not part of the user facing API, similar to the following

```
template <typename T, typename I >
concept ReserveAndAssignable =
```

```

    InputIterator<I> && requires (C &c, C const &cc, range_size_t<C> s, I i) {
        c.reserve(s);
        cc.capacity();
        { cc.capacity() } -> range_size_t<C>;
        c.assign(i, i);
    };

```

LWG made the case that there should be a customization point for this behavior and that the set of requirements should be specified.

- Do we want implementation to be able to reserve the container memory before assigning the range (this was always the intent)? Do we want to allow it for arbitrary (non standard) containers?
- Do we want to document the set of requirements of reservable container as a Named requirements, exposition only concept, concept in the std namespace?
- Do we want a customization point (similar to `disable_sized_range`)? (The author believe the answer to that should be no - the risk of ambiguity is extremely low, we don't have an opt out for Container - in fact we don't have a Container concept at all), and I don't think that all concepts should have an opt-out.

For the purpose of `ranges::to` we further require that the container be constructible from the extra arguments passed to `ranges::to`.

A conservative approach would be to let the design unchanged for now (implementers can do reserve + assign for standard types only), and revisit the question for non-standard containers at a later date.

The wording in this revision of this paper reflects that conservative approach.

Alternative designs

While we believe the range constructor based approach is the cleanest way to solve this problem, LEWG was interested in alternative design based on free functions

Range constructors

The original version of that paper proposed to add range constructors to all constructors. This proved to be unworkable because of `std::initializer_list`:

```

std::vector<int> foo = ....;
std::vector a{foo}; //constructs a std::vector<std::vector<int>>
std::vector b(foo); //would construct a std::vector<int>

```

Tagged range constructors

To solve the problem described above, it is possible to add a tag

```
std::vector<int> foo = ....;
std::vector a{std::from_range, foo}; //constructs a std::vector<int>
```

This will be explored in a separate paper by Tristan Brindle. However this approach does not replace `ranges::to`, which has the advantages of being a pipe adaptor and works with non-standard containers which do not support such tag constructors.

`ranges::to` can take advantages of these ranges constructors when available. The proposed wording in this paper body assumes that tagged constructors will be adopted in addition to the current paper.

Existing practices

Range V3

This proposal is based on the `to` (previously `to_`) function offered by ranges v3.

```
auto vec = view::ints
| view::transform([](int i) {
    return i + 42;
})
| view::take(10)
| to<std::vector>;
```

Abseil

Abseil offers converting constructors with each of their views. As per their documentation:

One of the more useful features of the `StrSplit()` API is its ability to adapt its result set to the desired return type. `StrSplit()` returned collections may contain `std::string`, `absl::string_view`, or any object that can be explicitly created from an `absl::string_view`. This pattern works for all standard STL containers including `std::vector`, `std::list`, `std::deque`, `std::set`, `std::multiset`, `std::map`, and `std::multimap`, and even `std::pair`, which is not actually a container.

Because they can not modify existing containers, view materialization in Abseil is done by the means of a conversion operator:

```
template<Container C>
operator C();
```

However, because it stands to reason to expect that there are many more views than containers and because conversions between containers are also useful, it is a more general solution to provide a solution that is not coupled with each individual view.

Previous work

[N3686] explores similar solutions and was discussed by LEWG long before the Ranges TS.

Proposed wording

Wording is relative to [?].

Add to the synopsis in [ranges.syn]:

```
namespace std::ranges {  
  
    template <input_range C, input_range R, typename... Args>  
    requires (!view<C>)  
    constexpr C to(R && r, Args&&...);  
  
    template <template <typename...> typename C, input_range R, typename... Args>  
    constexpr auto to(R && r, Args&&...) -> see below;  
  
}
```

In [range.utility] Insert after section [range.dangling]

Container conversions **[range.utility.container]**

The container conversions functions provide functions that efficiently convert ranges to containers. The container is constructed from the begin(range)/end(range) iterators pair.

```
template <input_range C, input_range R, typename... Args>  
requires (!view<C>)  
constexpr C to(R && r, Args&&... args);
```

Construct a container C with the elements of r in the following manner:

- If `std::constructible_from<C, R, Args...>` is true, equivalent to `C(std::forward<R>(r), std::forward<Args>(args)...) .`
- Otherwise, if `std::constructible_from<C, from_range_t, R, Args...>` is true, equivalent to `C(from_range, std::forward<R>(r), std::forward<Args>(args)...) .`
- Otherwise, if `std::constructible_from<C, Args...>` is true and `std::indirectly_copyable<ranges::range_iterator_t<R>, ranges::range_iterator_t<C>>` is true, equivalent to:

```
C c(std::forward<Args...>(args)...);  
ranges::copy(std::forward<R>(r), std::back_inserter(c));
```
- Otherwise, if:
 - `ranges::input_range<ranges::range_value_t<C>>` is true and
 - `ranges::input_range<ranges::range_value_t<R>>` is true and
 - `ranges::view<ranges::range_value_t<C>>` is false and


```
- std::indirectly_copyable<
  ranges::range_iterator_t<ranges::range_value_t<R>>,
  ranges::range_iterator_t<ranges::range_value_t<C>>
> is true
```

equivalent to:

```
C c(std::forward<Args...>(args)...);
auto v = r | transform ([](auto && elem) {
    return to<range_value_t<C>>(elem);
});
ranges::copy(v, std::back_inserter(c));
```

- Otherwise `ranges::to<C>(r, args)` is ill-formed.

```
template <template <typename...> typename C, input_range R, typename... Args>
constexpr auto to(R && r, Args&&... args) -> ContainerType;
```

Equivalent to `ranges::to<ContainerType>(std::forward<R>(r), std::forward<Args>(args)...) where ContainerType is determined using CTAD as follow:`

Let *ITER* be a type meeting the requirements of Cpp17InputIterator such that

- `same_as<ITER::iterator_category, input_iterator_tag>` is true,
- `same_as<ITER::value_type, ranges::range_value_t<R>>` is true,
- `same_as<ITER::difference_type, ranges::range_difference_t<R>>` is true,
- `same_as<ITER::pointer, ranges::range_value_t<R>*>` is true,
- `same_as<ITER::reference, ranges::range_reference_t<Rng>>` is true.

Then, *ContainerType* is:

- `decltype(C(std::declval<R>(), std::declval<Args>()...))` if that is a valid expression,
- Otherwise, `decltype(C(from_range, std::declval<R>(), std::declval<Args>()...))` if that is a valid expression,
- Otherwise, `decltype(C(std::declval<ITER>(), std::declval<ITER>(), std::declval<Args>()...))` if that is a valid expression,
- Otherwise `ranges::to<C>(r, args)` is ill-formed.



ranges::to adaptors

[range.utility.container.adaptors]

In addition to the functions described above, `ranges::to` also defines a closure object that accepts a `viewable_range` argument and returns a *Container* such that the expressions `r | ranges::to<Container>(args...)` and `ranges::to<Container>(r, args...)` have equivalent semantics. [Note: *Container* denotes either a class or a class template — end note].

The bitwise OR operator is overloaded for the purpose of chaining `ranges::to` to the end of an adaptor chain pipeline.

[*Example:*

```
list<int> ints{0,1,2,3,4,5};

auto v1 = ints | ranges::to<vector>();
auto v2 = ints | ranges::to<vector<int>>>();
auto v3 = ranges::to<vector>(ints);
auto v4 = ranges::to<vector<int>>>(ints);

assert(v1 == v2 && v2 == v3 && v3 == v4);
```

— *end example*]

Implementation Experience

Implementations of this proposal are available on in the 1.0 branch of [RangeV3] and in [cmcstl2]. Another implementation reflecting exactly this proposal is available on Github [rangesnext].

Related Paper and future work

- P1989R0 [?] adds range and iterator constructor to `string_view`
- [P1425] adds iterator constructors to `stack` and `queue`
- [P1419] Provide facilities to implementing span constructors more easily.

Future work is needed to allow constructing `std::array` from *tiny-ranges*.

Acknowledgements

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References

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- [RangeV3] Eric Niebler https://github.com/ericniebler/range-v3/blob/v1.0-beta/include/range/v3/to_container.hpp

- [rangesnext] Corentin Jabot <https://github.com/cor3ntin/rangesnext/blob/master/include/cor3ntin/rangesnext/to.hpp>
- [CTAD Ranges] Eric Niebler <https://github.com/ericniebler/range-v3/blob/d284e9c84ff69bb416d9d94d029729dfb38c3364/include/range/v3/range/conversion.hpp#L140-L152>
- [P1391] Corentin Jabot *Range constructor for std::string_view*
<https://wg21.link/P1391>
- [P1394] Casey Carter, Corentin Jabot *Range constructor for std::span*
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- [P1425] Corentin Jabot *Iterators pair constructors for stack and queue*
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- [N3686] Jeffrey Yasskin *[Ranges] Traversable arguments for container constructors and methods*
<https://wg21.link/n3686>
- [P1072R1] Chris Kennelly, Mark Zeren *Vector as allocation transfer device* <https://wg21.link/P1072>
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